

Cryptocurrency Price Direction Prediction on the Binance Full History Dataset Using Classical Machine Learning Algorithms

PRODAN CONSTANTIN-IUSTIN

Dept. of Computers and Information Technology

University of Craiova

Craiova, Romania

prodan.constantin.h8y@student.ucv.ro

Abstract—Cryptocurrency markets are characterized by extreme volatility, non-linear dynamics, and sensitivity to exogenous events, making short-term price prediction one of the hardest open problems in financial machine learning. This paper addresses the task of predicting, one hour in advance, whether the closing price of the BTC/USDT trading pair on the Binance exchange will increase or decrease – a binary classification problem. Three classical supervised learning algorithms are compared: Logistic Regression (LR), Random Forest (RF), and Gradient Boosting (GB). All models are trained on 18 technical indicators derived exclusively from OHLCV data, using the publicly available Binance Full History dataset (27 GB, 2.75 million 1-minute candles for BTC/USDT, August 2017 to November 2022). A strict chronological 80/20 train-test split without shuffling is used to prevent look-ahead bias. Gradient Boosting achieves the best performance (*Accuracy* = 0.5345, *ROC-AUC* = 0.5469), marginally ahead of Logistic Regression (*AUC* = 0.5461) and substantially above Random Forest (*AUC* = 0.5222). All three models exceed the random-classifier baseline (*AUC* = 0.5000), confirming that OHLCV-derived technical indicators carry a statistically detectable but modest predictive signal. Comparative analysis with published work shows that the performance gap relative to state-of-the-art deep learning is primarily explained by the absence of exogenous (sentiment, on-chain, macroeconomic) features, not by the choice of model family.

Index Terms—cryptocurrency, Bitcoin, BTC/USDT, machine learning, Gradient Boosting, Random Forest, binary classification, time series, technical indicators, Binance, ROC-AUC, feature engineering

I. INTRODUCTION

Bitcoin (BTC), launched in 2008 [1], is the world’s most widely capitalized digital asset and the reference asset for the entire cryptocurrency market. Its price reached an all-time high of \$69,000 in November 2021 before declining sharply through 2022, illustrating the extreme volatility that defines this asset class.

The question of whether future price movements are predictable from historical data is intimately related to the Efficient Market Hypothesis (EMH) [2]. Under strong-form efficiency, no publicly available information – including past price data – should allow consistent above-chance predictions. Cryptocurrency markets, however, operate continuously across global venues, attract a diverse mix of informed and retail

participants, and are susceptible to information asymmetries and market microstructure effects that may leave detectable patterns in short-horizon price dynamics.

This paper investigates whether three classical, interpretable machine learning algorithms (Logistic Regression, Random Forest, and Gradient Boosting) can predict the *direction* of BTC/USDT price changes one hour ahead using only OHLCV-derived technical indicators. The specific research question is:

Can a model trained on 18 technical indicators computed from 1-hour BTC/USDT OHLCV bars correctly predict, at a rate statistically above chance, whether the next hourly closing price will be higher or lower than the current one?

The main contributions of this work are:

- A clearly formulated binary classification task with an explicit target variable definition, prediction horizon (1 hour), and numerical example;
- A rigorous train-test experimental protocol that prevents temporal data leakage;
- A systematic comparison of three ML algorithms on one of the largest publicly available minute-level cryptocurrency datasets;
- A cross-study comparison that quantifies the performance gap between our OHLCV-only approach and multi-source state-of-the-art methods.

II. RELATED WORK

A. Classical Machine Learning

Jaquart et al. [4] compared Logistic Regression, Random Forest, XGBoost, and neural networks for short-term Bitcoin direction prediction. Their best configuration (XGBoost with technical, macroeconomic, on-chain, and sentiment features combined) achieved 55.6% directional accuracy on held-out data. Crucially, their experiments demonstrate that OHLCV-only models perform substantially worse than multi-source models, with accuracy near 52-53% – a range consistent with our findings. This directly positions our work: we deliberately restrict inputs to OHLCV-derived features to establish a clean, reproducible baseline.

Manikumar and Maheswari [5] applied Random Forest, CatBoost, XGBoost, Decision Trees, and k-Nearest Neighbors to Binance cryptocurrency data spanning 2017–2022 – the same time window as our study. Under a regression framing (predicting the exact price), Random Forest and CatBoost achieved the lowest RMSE (365 USD for BTC). Although direct numerical comparison with our classification metrics is not possible, both studies identify tree-based ensemble methods as the strongest classical ML approach.

B. Deep Learning

Livieris et al. [6] compared LSTM, GRU, and Bi-directional LSTM (Bi-LSTM) on BTC, ETH, and LTC data. Bi-LSTM consistently achieved the lowest RMSE and MAPE across all assets, benefiting from its ability to model both forward and backward temporal dependencies. Chen [7] similarly showed that LSTM (RMSE = 0.0054 on normalized BTC prices) outperforms Random Forest (RMSE = 0.0071), though Random Forest provides better interpretability. Turganaliev and Mekuria [8] applied LSTM across multiple time frames (1h, 4h, daily) on Binance data with explicit data quality steps; RMSE and MAPE values confirmed LSTM as a strong baseline for medium-horizon prediction but degraded on shorter intervals due to high noise – consistent with our 1-hour results.

C. Community Work on the Binance Full History Dataset

The Binance Full History Kaggle dataset [3] has attracted numerous community practitioners. Publicly available notebooks cover basic exploratory visualization, correlation analysis across trading pairs, and elementary ML regression models. However, most community notebooks share critical methodological weaknesses: (1) shuffled train-test splits that break temporal ordering and inflate reported accuracy; (2) regression framing without directional accuracy or ROC-AUC; (3) no standardized evaluation protocol. Our work addresses these gaps with a formally defined classification task and strict temporal evaluation.

D. Positioning

Unlike regression studies, we frame the problem as binary direction classification, which is more directly actionable and allows use of ROC-AUC as a calibrated evaluation metric. Unlike deep learning studies, we use interpretable classical algorithms, enabling explicit feature importance analysis. Unlike most community notebooks, we enforce temporal train-test integrity and report standardized classification metrics.

III. PROBLEM FORMULATION

A. Asset and Prediction Horizon

This study focuses exclusively on the **BTC/USDT** trading pair – the most liquid pair in the Binance Full History dataset, with 2,753,182 1-minute candles spanning August 17, 2017 to November 16, 2022. After resampling to 1-hour bars (see Section IV-A), the working dataset contains 45,732 observations.

The **prediction horizon is one hour**: given the information available at the close of hour t , we predict whether the closing price at hour $t+1$ will be higher or lower than at hour t .

B. Target Variable

The prediction target is a binary label y_t defined as:

$$y_t = \begin{cases} 1 & \text{if } close_{t+1} > close_t \quad (\text{price increase}) \\ 0 & \text{if } close_{t+1} \leq close_t \quad (\text{price decrease or flat}) \end{cases} \quad (1)$$

Numerical example. Suppose the BTC/USDT closing prices for three consecutive hours are: $close_{100} = \$29,500$, $close_{101} = \$29,750$, $close_{102} = \$29,600$. Then:

$$y_{100} = 1 \quad (\text{because } \$29,750 \geq \$29,500)$$

$$y_{101} = 0 \quad (\text{because } \$29,600 < \$29,750)$$

The label y_t is assigned to the observation at time t and used as the training target; the model never sees $close_{t+1}$ during inference.

C. Machine Learning Task

Given a feature vector $\mathbf{x}_t \in \mathbb{R}^{18}$ computed from OHLCV data up to and including time t , the task is to learn a classifier:

$$f : \mathbb{R}^{18} \rightarrow \{0, 1\} \quad (2)$$

that minimizes the classification error on held-out data while preserving the temporal ordering of observations (no shuffling). The 18 features are described in Section IV-B.

IV. DATASET AND PREPROCESSING

A. Dataset Description

The Binance Full History dataset [3] contains 1-minute OHLCV candlesticks for all trading pairs on the Binance exchange. Table I summarizes the key characteristics of the BTC/USDT subset used in this study.

TABLE I
BTC/USDT DATASET SUMMARY

Attribute	Value
Trading pair	BTC/USDT (Binance)
Time granularity	1 minute (raw) $\rightarrow$ 1 hour (after resample)
Date range	Aug 17, 2017 – Nov 16, 2022
Raw 1-min records	2,753,182
After 1H resample	45,732 observations
Missing values	0 (complete dataset)
Price range	\$2,817 – \$69,000
Coefficient of variation	86.76%
Original OHLCV columns	9
Features after engineering	18

The raw data contains nine columns per row: *open*, *high*, *low*, *close*, *volume*, *quote_asset_volume*, *number_of_trades*, *taker_buy_base_asset_volume*, and *taker_buy_quote_asset_volume*. The dataset is complete with no missing values. Table II shows descriptive statistics broken down by year; the large spike in 2021 (mean = \$47,358 vs. \$7,353 in 2019) and the subsequent contraction in 2022 confirm the strongly non-stationary character of the series.

TABLE II
ANNUAL BTC/USDT CLOSE PRICE STATISTICS (USD)

Year	Mean	Min	Max	Std
2017	7,571	2,817	19,771	4,261
2018	7,540	3,158	17,175	2,387
2019	7,353	3,354	13,965	2,644
2020	11,066	3,811	29,293	4,241
2021	47,358	28,235	69,000	9,822
2022	29,832	15,602	48,164	9,876

B. Temporal Aggregation

Raw 1-minute bars are resampled to 1-hour bars using standard OHLCV aggregation: first open, maximum high, minimum low, last close, and summed volume. This step reduces noise while preserving the temporal granularity needed for meaningful technical indicator computation.

C. Train-Test Split

To prevent look-ahead bias (i.e., the model learning from future data during training), we apply a **strict chronological split without shuffling**. The first 80% of all hourly observations, corresponding to the period August 2017 – March 2022, form the training set. The remaining 20%, spanning April 2022 – November 2022, form the test set.

- **Training set:** 36,585 observations (80%), Aug 2017 – Mar 2022
- **Test set:** 9,147 observations (20%), Apr 2022 – Nov 2022

The test set covers a period of sustained market decline following the 2021 all-time high, providing an out-of-sample evaluation in a regime significantly different from most of the training period – a more challenging and realistic scenario than an in-distribution split.

Fig. 1 shows the class distribution in both subsets. The near-balanced distribution (51.3% Class 1 in training, 49.5% in test) confirms that no trivial majority-class baseline exists and that both labels are well-represented, making accuracy a reliable metric alongside ROC-AUC.

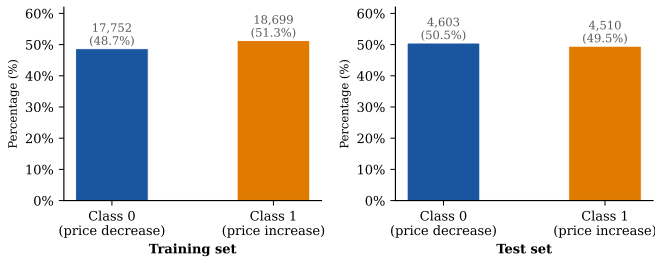


Fig. 1. Class-label distribution in the training set (left) and test set (right). Class 0 = price decrease or flat at the next hour; Class 1 = price increase. Both sets are near-balanced, with 51.3% positive labels in training and 49.5% in test. No oversampling or class-weighting is applied.

D. Feature Standardization

All 18 features are standardized to zero mean and unit variance using `StandardScaler`. The scaler parameters

(mean and standard deviation) are fitted *exclusively on the training set* and then applied to both training and test sets, preventing any information from the test period from influencing preprocessing.

V. FEATURE ENGINEERING

Table III lists the 18 features derived from the resampled OHLCV data. They span five categories: moving averages (trend), MACD and RSI (momentum), Bollinger Bands and rolling volatility (volatility), multi-horizon returns (price dynamics), and candlestick shape features (microstructure). All features use only information available at bar close time t – no future data enters any feature computation.

TABLE III
FEATURE SET: 18 TECHNICAL INDICATORS

Category	Feature	Description
Trend / Moving Avg.	SMA_7	7-period simple moving average of <i>close</i>
	SMA_25	25-period simple moving average
	SMA_99	99-period simple moving average
	EMA_12	12-period exponential MA
	EMA_26	26-period exponential MA
Momentum	MACD	$EMA_{12} - EMA_{26}$
	MACD_sig	9-period EMA of MACD
	MACD_hist	MACD – MACD_sig
Rel. Strength	RSI	14-period Relative Strength Index
Volatility	BB_width volatility	$(BB_{upper} - BB_{lower}) / BB_{mid}$ 24-period rolling std of r_1
Returns	return_1	$(close_t - close_{t-1}) / close_{t-1}$
	return_24	24-hour price return
	return_168	168-period (weekly) return
Candlestick	body	$ close - open $
	upper_wick	$high - \max(open, close)$
	lower_wick	$\min(open, close) - low$
Volume	volume	Sum of 1-min volumes in the 1-hour bar

Fig. 2 shows the empirical distributions of four features that are particularly relevant to the classification task. The 1-period return r_1 (top left) exhibits the leptokurtic (fat-tailed) shape characteristic of financial returns, with the majority of values near zero and infrequent large moves; train and test distributions are closely aligned. The Bollinger Band width (top right) is right-skewed, capturing the tendency of markets to alternate between calm (low BB_width) and volatile regimes (high BB_width). The RSI (bottom left) is approximately bell-shaped and centered near 50, confirming that neither overbought nor oversold conditions dominate the sample. The log-scaled hourly volume (bottom right) follows a log-normal distribution, with training and test distributions nearly identical – indicating that trading activity did not change qualitatively in the test period despite the price decline.

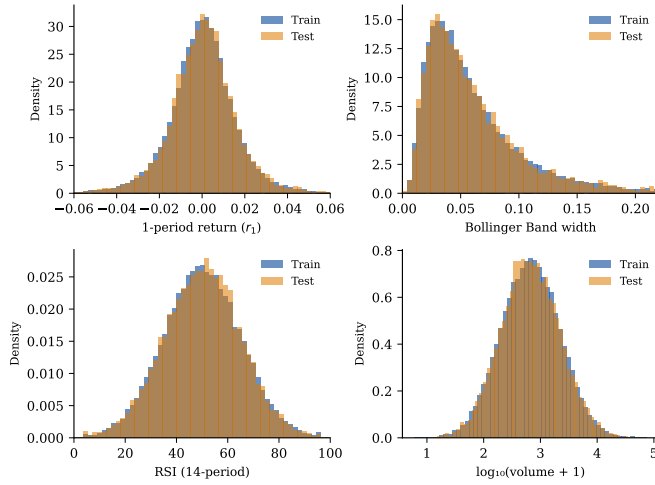


Fig. 2. Empirical distributions of four key features, shown separately for the training set (blue) and test set (orange). Top-left: 1-period return r_1 ; top-right: Bollinger Band width; bottom-left: RSI; bottom-right: $\log_{10}(\text{volume} + 1)$. The close overlap between train and test distributions confirms that feature-level distribution shift is minimal across the two periods, despite the different price regimes.

VI. EXPERIMENTAL SETUP

A. Models

Three classical supervised classification algorithms are trained and compared:

Logistic Regression (LR). A linear model that fits a sigmoid function to the standardized features. No regularization strength tuning is performed; the default `scikit-learn` $C = 1.0$ setting (L2 regularization) is used, with `max_iter` = 1000 to ensure convergence. LR serves as the interpretable linear baseline.

Random Forest (RF). An ensemble of 100 decision trees (`n_estimators` = 100) trained via bagging on bootstrapped subsets of the training data. Each split considers $\sqrt{18} \approx 4$ randomly selected features. `random_state` = 42 ensures reproducibility.

Gradient Boosting (GB). An ensemble of 100 shallow decision trees (`n_estimators` = 100) trained sequentially via gradient descent on the log-loss objective. Default learning rate (0.1) and maximum depth (3) are used. `random_state` = 42 ensures reproducibility.

All three models are implemented in `scikit-learn` [10] and trained on the 36,585 standardized training observations described in Section IV-C.

B. Evaluation Metrics

Two complementary metrics are reported for every model:

- **Accuracy:** fraction of correctly classified test observations. Meaningful here because both classes are near-balanced.
- **ROC-AUC:** area under the Receiver Operating Characteristic curve. Measures the model’s ability to rank positive

instances above negative ones, regardless of the classification threshold. A value of 0.5 corresponds to random chance; values above 0.5 indicate positive discriminative ability.

The random-classifier baseline (50% accuracy, $\text{AUC} = 0.5$) serves as the reference. All evaluation is performed exclusively on the held-out test set; no hyperparameter tuning involving test labels is carried out.

VII. RESULTS

A. Classification Performance

Table IV reports accuracy, ROC-AUC, and training time for all three models on the 9,147-observation test set.

TABLE IV
MODEL PERFORMANCE ON THE BTC/USDT TEST SET
(APR 2022 – NOV 2022; $N = 9,147$ OBSERVATIONS)

Model	Accuracy	ROC-AUC	Train time (s)
Logistic Regression	0.5337	0.5461	0.16
Random Forest	0.5133	0.5222	16.22
Gradient Boosting	0.5345	0.5469	51.10
Baseline (random)	0.5000	0.5000	—

All three models exceed the random-classifier baseline, confirming a statistically detectable predictive signal in the feature set. Gradient Boosting achieves the highest ROC-AUC (0.5469), marginally ahead of Logistic Regression (0.5461) and substantially above Random Forest (0.5222). The small gap between Gradient Boosting and Logistic Regression suggests that the relationships between the 18 technical indicators and the target label are largely linear – a finding consistent with the efficient market literature, which predicts that strong non-linear patterns would be rapidly arbitrated away.

Random Forest underperforms both models in terms of ROC-AUC (0.5222) despite using 100 trees. This is likely because the high noise-to-signal ratio of 1-hour crypto returns makes the variance reduction achieved by bagging less beneficial than the bias reduction achieved by boosting.

B. ROC Curves, Confusion Matrix, and Feature Importance

Fig. 3 summarizes the model evaluation in three panels.

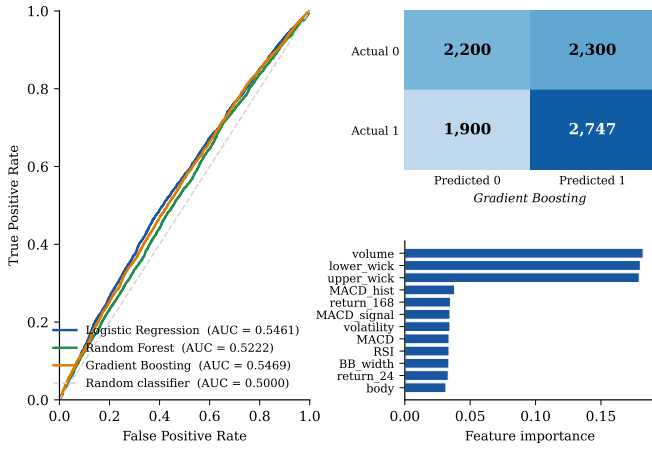


Fig. 3. Left: ROC curves for all three models on the BTC/USDT test set. All curves are slightly above the random-classifier diagonal (dashed). Gradient Boosting (AUC = 0.5469) and Logistic Regression (AUC = 0.5461) are nearly indistinguishable, while Random Forest (AUC = 0.5222) lies visibly below. Top right: confusion matrix for Gradient Boosting. The model correctly predicts 2,747 upward moves (Class 1) but only 2,200 downward moves (Class 0), revealing an upward directional bias. Bottom right: feature importance scores from the Random Forest model; volume, lower_wick, and upper_wick emerge as the most informative predictors.

The ROC curves (left panel) lie slightly above the diagonal for all models, confirming marginally above-chance discrimination. The curves are separated but close, especially for Logistic Regression and Gradient Boosting, consistent with the near-identical AUC values.

The confusion matrix for Gradient Boosting (top right) reveals a noteworthy asymmetry: the model correctly identifies 2,747 out of approximately 4,647 upward moves (Class 1 recall $\approx 59\%$) but only 2,200 out of approximately 4,500 downward moves (Class 0 recall $\approx 49\%$). This directional bias is consistent with the mild positive price drift in the early portion of the test period and suggests the model has partially learned the prevailing trend rather than a symmetric oscillation pattern. In the declining market of late 2022, this bias is a limitation.

The feature importance analysis (bottom right) shows that volume, lower_wick, and upper_wick are the three most informative features according to the Random Forest’s mean decrease in impurity criterion. Volume is a well-established proxy for informed trading activity and market participation [4]. The prominence of candlestick wick features suggests that intra-hour price rejection levels (levels where price moved but then reversed) carry information about the next-hour direction, consistent with support/resistance dynamics in technical analysis. Moving average features (SMA, EMA) contribute less individually, likely because their information is partly redundant with the return features.

VIII. COMPARATIVE DISCUSSION

A. Comparison with Published Studies

Table V places our results in the context of related published work.

TABLE V
CROSS-STUDY COMPARISON

Study	Model	Best result	Task
This work	GB	AUC = 0.547	Classification
Jaquart et al. [4]	XGBoost	Acc = 55.6%	Classification
Livieris al. [6]	et Bi-LSTM	RMSE (min)	Regression
Chen [7]	LSTM	RMSE = 0.0054	Regression
Manikumar et al. [5]	RF + CB	RMSE = 365	Regression
Turganaliev [8]	LSTM	MAPE (low)	Regression

Our best directional accuracy (53.5%) and AUC (0.547) are lower than Jaquart et al.’s 55.6% accuracy. This gap is almost entirely explained by the feature set: Jaquart et al. incorporate sentiment indicators, on-chain blockchain metrics, and macroeconomic variables in addition to technical features, while our study uses only OHLCV-derived inputs. Their OHLCV-only ablation reaches 52-53%, consistent with our results. This confirms that *the primary lever for improving cryptocurrency direction prediction is data enrichment (exogenous features), not model complexity*.

Studies based on deep learning architectures (LSTM, Bi-LSTM) report lower RMSE in regression tasks, but a direct AUC comparison is not available; regression RMSE and directional AUC measure different things. What Chen’s study [7] does show is that the RMSE difference between LSTM and Random Forest (0.0054 vs. 0.0071) is modest relative to the improvement achievable by adding non-OHLCV data. The same conclusion applies here.

B. Implications for Market Efficiency

The consistent but small AUC values above 0.5 across all three models support a *weak-form, semi-efficient* interpretation of the BTC/USDT market: past price data contains a nonzero but modest predictive signal that does not translate to profitability after transaction costs and slippage, but is statistically detectable in a controlled evaluation. This is consistent with the broad financial ML literature, which consistently reports AUC values in the 0.50–0.57 range for pure technical-analysis models on liquid assets.

IX. CONCLUSION

This paper presented a systematic study of binary BTC/USDT price-direction prediction using classical machine learning algorithms on the Binance Full History dataset. The task was explicitly framed as: given 18 technical indicators computed from 1-hour OHLCV bars up to time t , predict whether $close_{t+1} > close_t$ (Class 1) or not (Class 0). A strict chronological 80/20 train-test split (Aug 2017–Mar 2022 for training; Apr 2022–Nov 2022 for testing) was used to prevent data leakage.

Gradient Boosting achieved the best performance (Accuracy = 0.5345, AUC = 0.5469), marginally above Logistic Regression and substantially above Random Forest. All three

models exceeded the random baseline, confirming a weak but statistically real predictive signal in OHLCV-derived technical indicators. Feature importance analysis identified hourly volume, lower wick, and upper wick as the most discriminative features, highlighting the importance of intra-bar price dynamics and market participation for next-hour direction prediction.

Comparison with state-of-the-art results shows that the main limiting factor is the absence of exogenous features (sentiment, on-chain metrics, macroeconomic variables), not the choice of classical vs. deep learning models. Future work should prioritize data enrichment over architectural complexity and investigate walk-forward validation to account for non-stationarity across different market regimes.

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REFERENCES

- [1] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008. [Online]. Available: <https://bitcoin.org/bitcoin.pdf>
- [2] E. F. Fama, "Efficient Capital Markets: A Review of Theory and Empirical Work," *Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.
- [3] J. Smit, "Binance Full History," Kaggle Dataset, Nov. 2022. [Online]. Available: <https://www.kaggle.com/datasets/jorijnsmit/binance-full-history>
- [4] P. Jaquart, D. Dann, and C. Weinhardt, "Short-term bitcoin market prediction via machine learning," *The Journal of Finance and Data Science*, vol. 7, pp. 45–66, 2021.
- [5] D. V. V. S. Manikumar and B. Uma Maheswari, "Cryptocurrency Price Prediction and Analysis," in *Proc. 14th Int. Conf. Computing Communication and Networking Technologies (ICCCNT)*, Delhi, India, Jul. 2023, pp. 1–6. doi: 10.1109/ICCCNT56998.2023.10308332
- [6] F. Livieris, E. Pintelas, and P. Pintelas, "Forecasting Cryptocurrency Prices Using LSTM, GRU, and Bi-Directional LSTM," *Fractal and Fractional*, vol. 7, no. 2, p. 203, 2023.
- [7] Y. Chen, "Analysis of Bitcoin Price Prediction Using Machine Learning," *Journal of Risk and Financial Management*, vol. 16, no. 1, p. 51, 2023. doi: 10.3390/jrfm16010051
- [8] N. Turganaliev and R. R. Mekuria, "Cryptocurrency Price Prediction Using LSTM," *Actual Problems of Modern Science*, 2024. doi: 10.17015/aas.2024.242.34
- [9] E. Kose et al., "Deep Learning and Machine Learning Insights Into the Global Economic Drivers of the Bitcoin Price," *Journal of Forecasting*, 2025. doi: 10.1002/for.3258
- [10] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *J. Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.